

K-Nearest Neighborhood Formulae

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1 Distance Calculation

1. Euclidean distance

$$d(p, q) = \sqrt{\sum_{i=1}^n (p_i - q_i)^2}$$

2. Manhattan distance

$$d(p, q) = \sum_{i=1}^n |p_i - q_i|$$

3. Minkowski distance

$$d(p, q) = \left(\sum_{i=1}^n |p_i - q_i|^r \right)^{\frac{1}{r}}$$

2 Choosing K value

K is too small(K=1) -> Overfitting

K is too large(K=10) -> Underfitting

Usually start with 3,5,7,etc.

Use Cross-Validation to validate the K value

3 Voting or Averaging

Classification -> Majority Vote

Regression -> Average

formula:

$$\hat{y} = \frac{1}{k} \sum_{i=1}^k y_i$$